

Course Title: INTRODUCTION TO MECHANICAL ENGINEERING

Course Code: 22ESC144/244

Course Incharge: Ashoka E

Module-2

Machine Tool Operations:

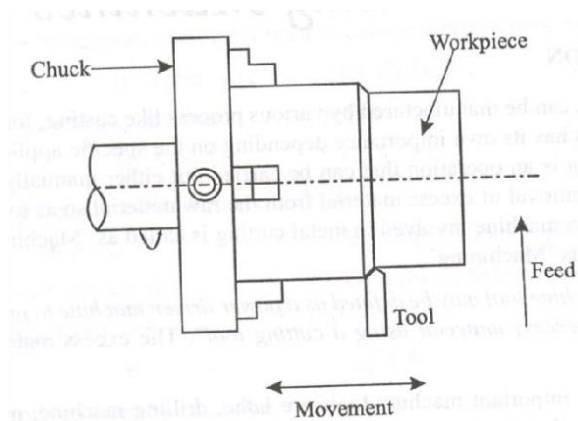
INTRODUCTION –

A product or part can be manufactured by various processes like casting, forging, machining etc. Machining is an operation that can be carried out manually or by machine, which involves *removal of excess material from the raw material* so as to get the required shape and size. So any machine involved in metal cutting is known as a ‘Machine tools’ and the process is called ‘Machining’.

“A machine tool may be defined as a power driven machine to produce a product by removing the excess material using a cutting tool”. The excess material removed is called as chips. Some of the important machine tools are lathe, drilling machine, milling machine, grinding machine, shaper etc.

Lathe :-A Lathe is a machine tool employed generally to produce circular object. It is said to be the mother of all machine tools, generally used to perform on other machine tool like grinding shaping milling can be performed

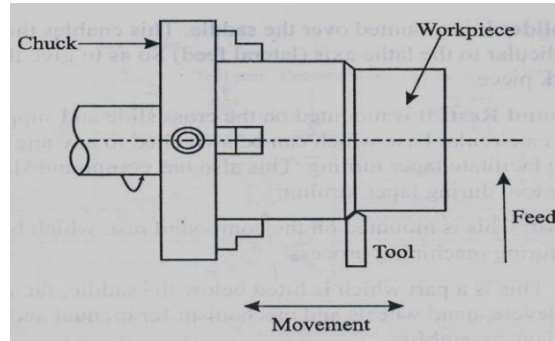
Principle of working



A lathe, basically a turning machine works on the principle that a cutting tool can remove material in the form of chips from the rotating work pieces to produce circular objects. This is accomplished in a lathe which holds the work pieces rigidly and rotates them at high speeds while a cutting tool is moved against it.

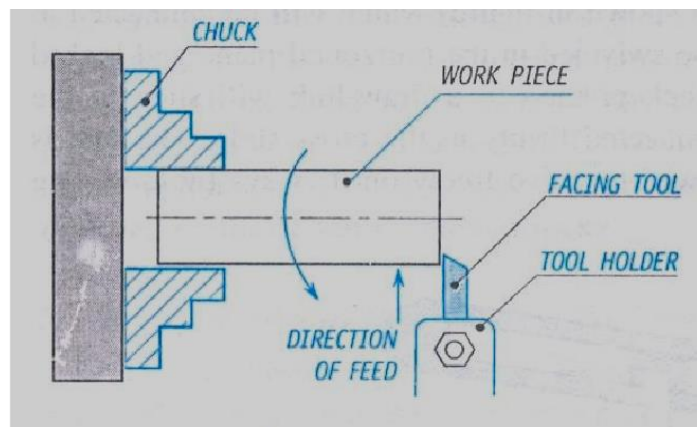
Operation on Lathe

1. Plain Turning (Turning)



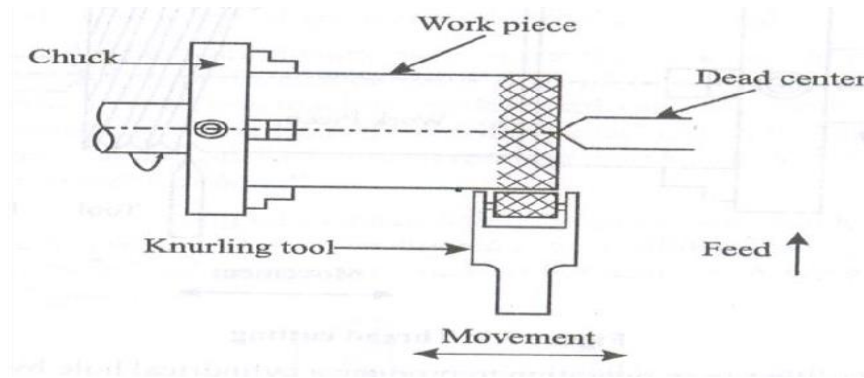
The work piece is supported between the two centers which permit the rotation of the work piece. The cutting tool is fed perpendicular to the axis of the work piece to a given depth of cut, and is then moved parallel to the axis of the work piece. “This method of machining operation in which the work piece is reduced to a cylindrical section of required diameter is called ‘turning’”.

2. Facing



This operation on lathe is used to produce either flat surface or shoulders at the end of the work piece and used to reduce the Length of the work piece. The work piece is held rigidly in the chuck. The cutting tool is fed perpendicular to the axis of the lathe from the outer edge of the work piece towards the center.

3.Knurling



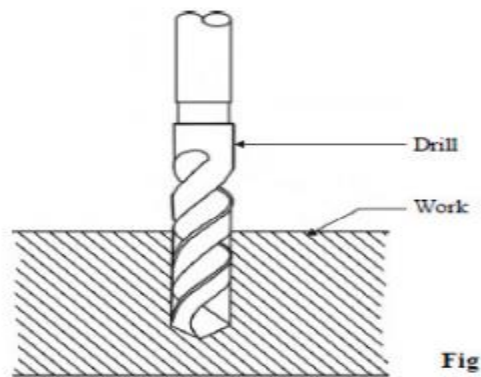
It is an operation used to emboss serrated surface on work piece using a knurling tool. It is used to produce straight, angled or diamond pattern on the work piece mainly for gripping purpose. The knurling tool is set in the tool post such that upper and lower rollers of the knurl head touch the surface of the work piece.

DRILLING MACHINE

Drilling machine is one of the most important machine tools in a workshop. It was designed to produce a cylindrical hole of required diameter and depth on metal workpieces. Though holes can be made by different machine tools in a shop, drilling machine is designed specifically to perform the operation of drilling and similar operations. Drilling can be done easily at a low cost in a shorter period of time in a drilling machine.

Drilling can be called as the operation of producing a cylindrical hole of required diameter and depth by removing metal by the rotating edges of a drill. The cutting tool known as drill is fitted into the spindle of the drilling machine. A mark of indentation is made at the required location with a centre punch. The rotating drill is pressed at the location and is fed into the work. The hole can be made upto a required depth

Working principle: The rotating edge of the drill exerts a large force on the workpiece and the hole is generated. The removal of metal in a drilling operation is by shearing and extrusion. A mark of indentation is made at the required location with a centre punch. The rotating drill is pressed at the location and is fed into the work. The hole can be made upto a required depth



Fig

Sensitive drilling machine

It is designed for drilling small holes at high speeds in light jobs. High speed and hand feed are necessary for drilling small holes.

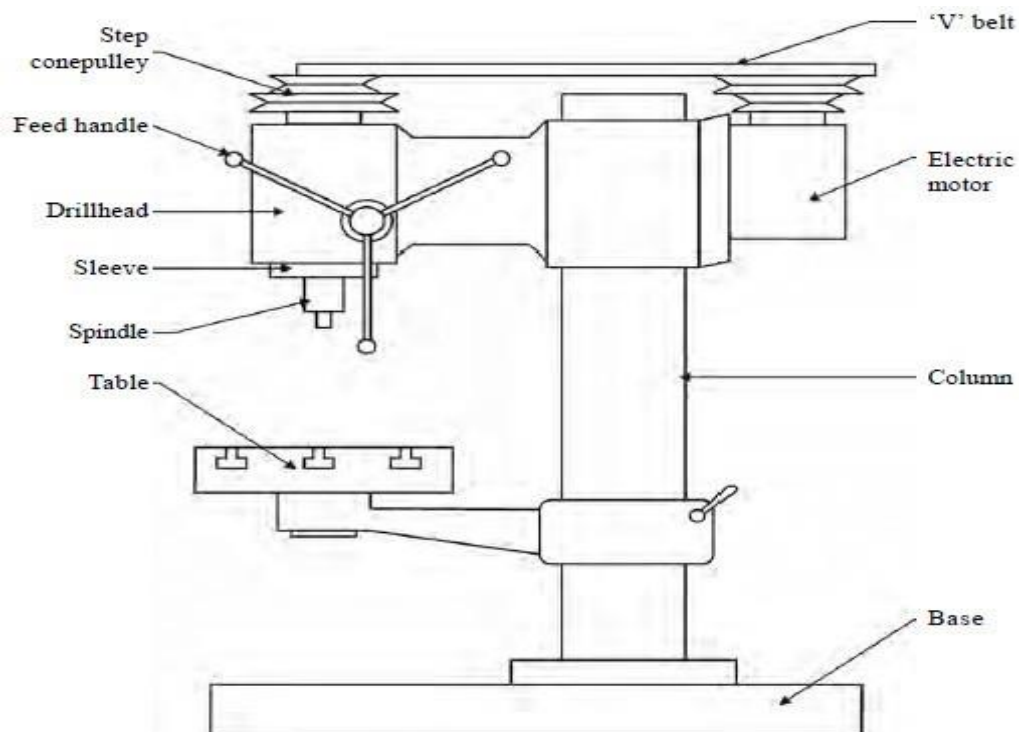


Fig 2.1 Sensitive drilling machine

The base of the machine is mounted either on a bench or on the floor by means of bolts and nuts. It can handle drills upto 15.5mm of diameter. The drill is fed into the work purely by hand. The operator can sense the progress of the drill into the work because of hand feed. The machine is named so because of this reason.

A sensitive drilling machine consists of a base, column, table, spindle, drill head and the driving mechanism.

A sensitive drilling machine is shown in Fig. 2.1.

Base

The base is made of cast iron and so can withstand vibrations. It may be mounted on a bench or on the floor. It supports all the other parts of the machine on it.

Column

The column stands vertically on the base at one end. It supports the work table and the drill head. The drill head has drill spindle and the driving motor on either side of the column.

Table

The table is mounted on the vertical column and can be adjusted up and down on it. The table has 'T'-slots on it for holding the workpieces or to hold any other work holding device.

Drill head is mounted on the top side of the column. The drill spindle and the driving motor are connected by means of a V-belt and cone pulleys. The motion is transmitted to the spindle from the motor by the belt.

Drilling

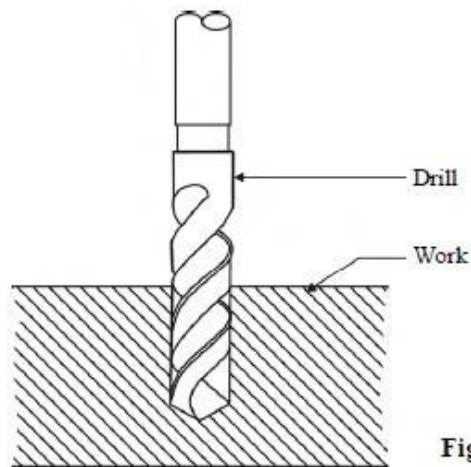


Fig 2.17 Drilling

Drilling is one of the machining operations of **producing circular hole in the work-piece by using a rotating cutter called drill.**

Reaming

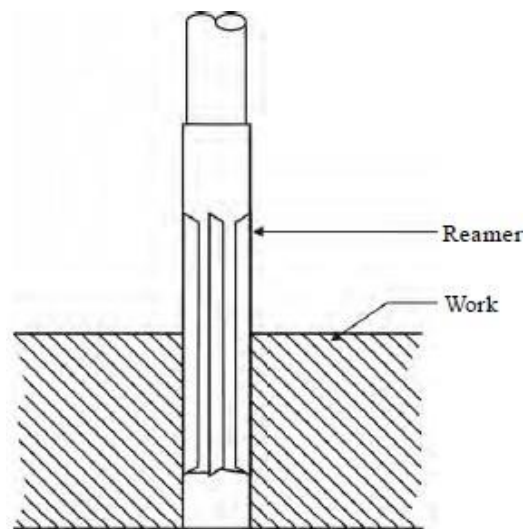


Fig 2.18 Reaming

Reaming is a cutting process that involves the use of a rotary cutting tool to create smooth interior walls in an existing hole in a workpiece. The rotary cutting tool used in reaming is known as a reamer. Like drill bits, reamers also remove material from the workpiece on which they are used

Boring

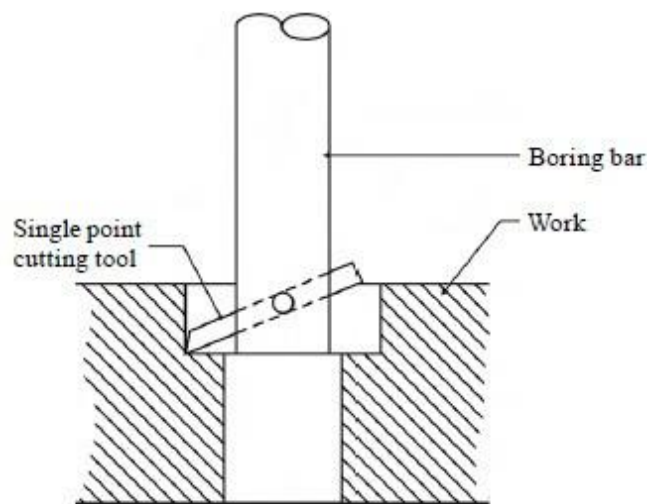


Fig 2.19 Boring

Boring is the operation of truing and enlarging a previously drilled hole by means of a single point cutting tool. Boring is done on drilling machine to perform the following tasks on a hole already drilled

Milling Machine

Milling is a metal removal process in which the work piece is fed into a revolving cutting tool, thereby removing excess material. The cutting tool is known as milling cutter which is a multipoint cutting tool. Milling Machine – It is a power operated machine tool in which the work piece mounted on the table is fed against the milling cutter to get the required shape.

PRINCIPLE OF MILLING

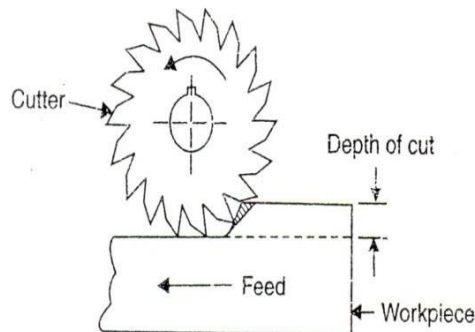
In milling, the work piece is rigidly held on table and is slowly fed into the uniformly rotating cutter to get the required shape.

The work piece can be fed in either direction of the rotating cutter.

(a) Up milling (conventional)

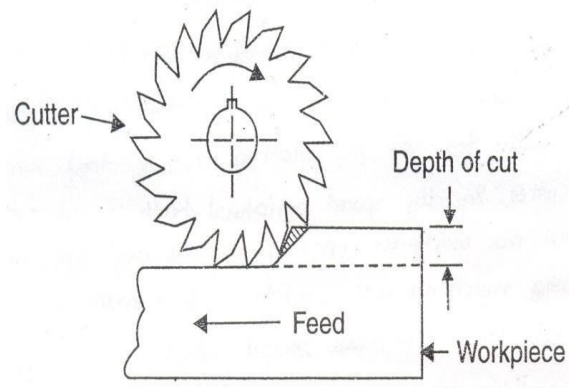
(b) Down milling (climb)

a).UP Milling (Conventional)



The work piece is fed opposite to the direction of the rotating cutter. Chips are progressively thicker. Since the cutting forces are directed upwards it tends to lift the work piece. Gives poor surface finish. Used for hard materials.

(b) Down Milling (Climb)



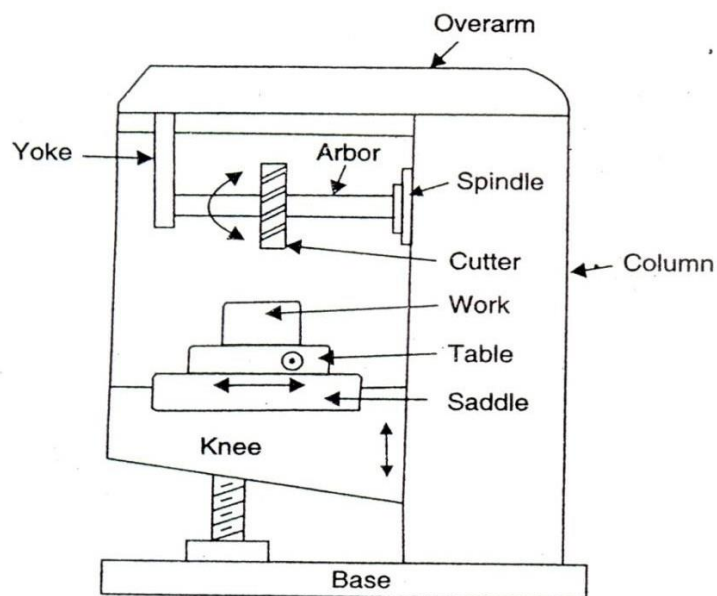
The work piece in the direction of the rotating cutter. Chips are progressively thinner. Cutting forces are directed downwards, which keep the work piece pressed to the table. Gives good surface finish. Used for soft materials and finishing operations.

Classification of Milling Machine

The milling machines are broadly classified into:

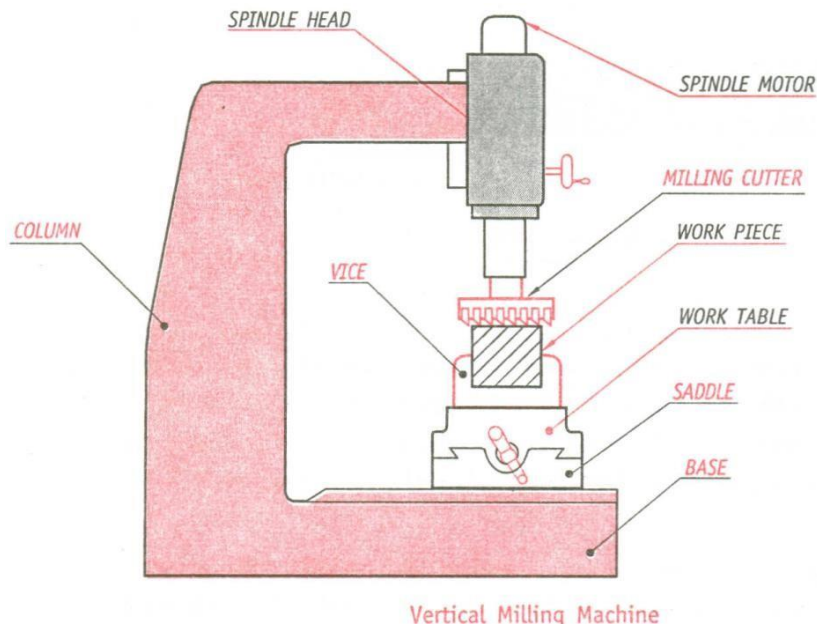
1. Plain or Horizontal Type of Milling Machine
2. Vertical Milling Machine

1. Horizontal Milling Machine (Column and knee type)



The horizontal milling machine shown in Fig. has its cutter axis horizontal. The work piece is held in a vice mounted on the machine table which is fitted over a saddle. The feed is given by moving the table against the horizontal axis of the revolving cutter. The cross feed handle enables the entire table to move across the knee. The knee can be raised or lowered by the knee elevating handle. This type of milling machine is commonly employed for milling operations which can be performed by feeding the workpiece in a straight line, either vertically or horizontally. Few examples of milling operations carried on this type of machine are: keyways, grooves, gear teeth, spline shafts, etc.

2. Vertical Milling Machine:



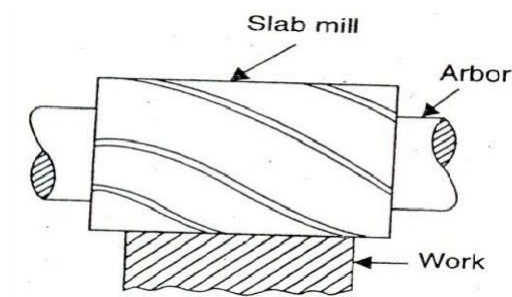
In a vertical milling machine shown in Fig, the spindle is mounted with its axis vertical perpendicular to the work table. The column and the base are formed into an integral casting. The spindle head is fitted vertically in the guideways provided in the projecting end of the column. The spindle can be moved up and down over the guideways. A saddle is mounted over the guideways provided on top of the base. The saddle can be moved in the transverse direction. The work table will be mounted over the saddle and can be moved longitudinally. In this machine, unlike the horizontal milling machine, the workpiece can be moved only in the horizontal plane both longitudinally and in the transverse direction, but not vertically.

Milling Process

1. Plain Milling or slab Milling

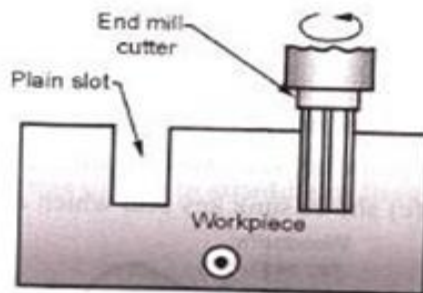
Plain milling is used for machining a flat, plain, horizontal surface, parallel to the axis of cutter with the help of plain or slab milling cutter. For machining of very wide surface, interlocking teeth plain milling cutters are used instead of simple plain milling cutters. During the operation, the work piece and cutter are secured properly on the machine and by rotating the vertical feed screw of the table and machine, the depth of cut is adjusted.

In plain milling, the surface of the workpiece is parallel to the



2. Slot Milling

Slot Milling Slot Milling is the process of milling slots using a different type of cutter called "Slot drill" which has the capacity to cut into solid material. Slot drill is used majorly in cases where it takes a lot of time to pre-drill a hole for an end mill and there is not enough room for the end mill to plunge using a helical motion.



COMPUTER NUMERICAL CONTROL (CNC)

What is CNC Machine

When computers are used to control a Numerical Control (NC) machine tool than the machine is called CNC machine. In other words, the use of computers to control machine tools like lathe, mills, slotter, shaper etc is called CNC machine. The cutting operations performed by the CNC is called CNC machining

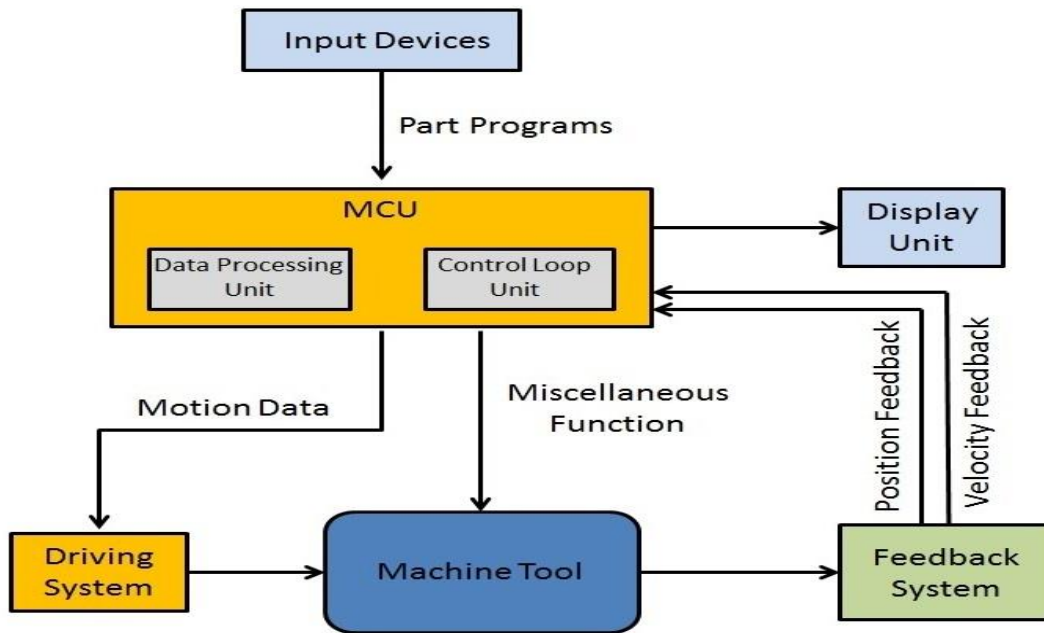
The Various machining services in which CNC machine is used are known as CNC machining services. In CNC machine shops, programs are designed or prepared first, and then it is fed to the CNC machine. According to the program, the CNC controls the motion and speed of the machine tools.



Fig: CNC Machine

CNC System consists of the following elements:

- i. Input Device
- ii. MCU or Machine Control Unit
- iii Machine Tool
- iv. Driving system
- v. System
- vi. Feedback devices
- vii. Display Unit



Main Parts of CNC Machine or Block diagram of CNC

The main parts of the CNC machine are

(i) Input Devices: These are the devices which are used to input the part program in the CNC machine. There are three commonly used input devices and these are punch tape reader, magnetic tape reader and computer via RS-232-C communication.

(ii) Machine Control Unit (MCU): It is the heart of the CNC machine. It performs all the controlling action of the CNC machine, the various functions performed by the MCU are

- It reads the coded instructions fed into it.
- It decodes the coded instruction.
- It feeds the axis motion commands to the amplifier circuits for driving the axis mechanisms.
- It receives the feedback signals of position and speed for each drive axis.
- It implements the auxiliary control functions such as coolant or spindle on/off and tool change.

(iii) Machine Tool: A CNC machine tool always has a slide table and a spindle to control of the position and speed. The machine table is controlled in X and Y axis direction and the spindle is controlled in the Z axis direction.

(iv) Driving System: The driving system of a CNC machine consists of amplifier circuits, drive motors and ball lead screw. The MCU feeds the signals (i.e. of position and speed) of each axis to the amplifier circuits.

(v) Feedback System: This system consists of transducers that act as sensors. It is also called a measuring system. It contains position and speed transducers that continuously monitor the position and speed of the cutting tool located at any instant. The MCU receives the signals from these transducers and it uses the difference between the reference signals and feedback signals to generate the control signals for correcting the position and speed errors.

(vi) Display Unit: A monitor is used to display the programs, commands and other useful data of CNC machine.

How CNC Machine Works.

- First, the part program is inserted into the MCU of the CNC.
- In MCU all the data process takes place and according to the program prepared, it prepares all the motion commands and sends it to the driving system.
- The drive system works as the motion commands are sent by MCU. The drive system controls the motion and velocity of the machine tool.
- The feedback system records the position and velocity measurement of the machine tool and sends a feedback signal to the MCU.
- In MCU, the feedback signals are compared with the reference signals and if there are errors, it corrects it and sends new signals to the machine tool for the right operation to happen.
- A display unit is used to see all the commands, programs and other important data. It acts as the eye of the machine

Advantages

- It can produce jobs with the highest accuracy and precision than any other manual machine.
- Fewer workers are required to operate a CNC and save labor costs.
It can be run for 24 hours a day.
- The parts produced by it have the same accuracy. There is no variation in the parts manufactured.
- Operators can easily make changes and improvements and reduce the delay time.
- It has the capability to produce complex designs with high accuracy in minimum possible time.
- The modern design software, allows the designer to simulate the manufacturer of his/her idea. And this removes the need for making a prototype or model and saves time and money.

Disadvantages

- The cost of the CNC machine is very high as compared with a manually operated machine.
- The parts of the CNC machines are expensive.
- The maintenance cost in the case of CNC is quite high.
- It does not eliminate the need for costly tools

Application

Almost every manufacturing industry uses CNC machines. With an increase in the competitive environment and demands, the demand for CNC usage has increased to a greater extent. The machine tools that come with the CNC are lathe, mills, shaper, welding, etc. The industries that are using CNC machines are the automotive industry, metal removing industries, industries of fabricating metals, electrical discharge machining industries, wood industries, etc.

Additive Manufacturing or 3D Printing

3D printing is the process of creating a three dimensional object from a digital design by building up layers of material with a 3D printer. 3D printing is also known as additive manufacturing because material is added during the process of manufacturing to form the object. The opposite of this, known as subtractive manufacturing, is where material is cut away from a sheet or a block of material to form the object, for example manufacturing processes such as milling or laser cutting.

Additive manufacturing wastes very little material compared to subtractive manufacturing. The way that 3D printing forms an object in layers allows for objects to be created with very complex geometries that simply could not be manufactured in any other way.

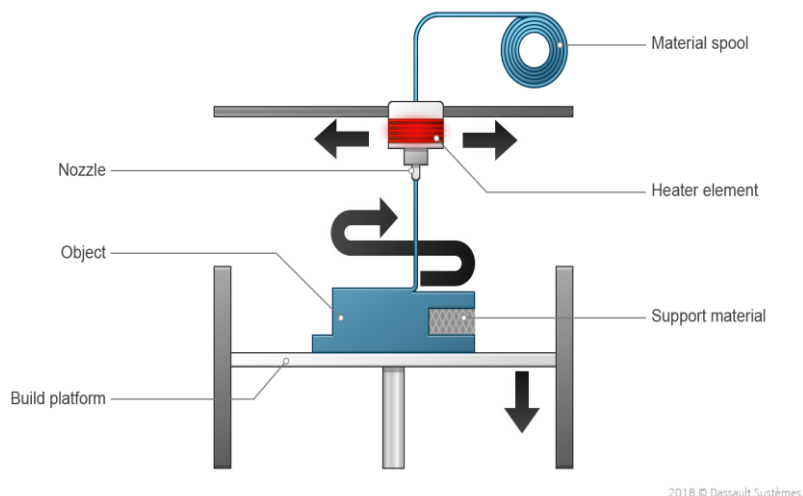


Fig: 3D Printing Machine

When the first layer is complete, the print bed moves down and the process is repeated for the second layer. The second layer of plastic is deposited on top of the -first layer and fans at either side of the hot end blow cold air over the printed plastic to cool it down. As the plastic cools the layers stick (or fuse) together. This process is repeated for each layer until the object is printed

Application of 3D printing

3D printing is now being used across a wide range of industries including engineering, automotive, aerospace, construction, architecture, medicine, product design, food and fashion.

Examples include:

- Used for rapid prototyping of parts and components in engineering, aerospace and automotive industry.
- In medicine, 3D printing from patient scans to assist in planning operations or for student training.
- Creation of custom jigs and templates for the automotive industry.
- Production of custom low cost prosthetics and disability aids.
- Small enterprises creating personalized 3D printed products and gifts.
- 3D printing shelters and humanitarian aids